Problem statement

Complete a partially observed bundle.

$$\underbrace{S = \{\text{shirt, trousers}\}}_{\text{observed items}} \quad \longrightarrow \quad \underbrace{c \in \mathcal{I} \setminus S}_{\text{candidate completion}}$$

$$\boxed{\widehat{C}_K(S) = \mathrm{TopK}_{c \notin S}\ \theta(c \mid S)}$$

$\mathcal{I}$: item catalogue. K: number of recommendations.

Goal: rank items that belong with the observed combination.

Problem statement

Specific historical evidence

These items were selected together.

Limitation: some candidates have no recorded matches.

Example: a new shoe has no recorded co-occurrence with this shirt or these trousers.

Transferable descriptive evidence

These items have related attributes.

Limitation: resemblance need not imply compatibility.

Example: trousers resembling the observed trousers may rank highly, even when the missing item is a shoe.

Research question

Can recorded bundle occurrences and item attributes support accurate completion without gradient-training a scoring model?

1. **Use observed combinations**
   Distinguish several query items occurring together from separate pairwise matches.
2. **Handle missing item-pair observations**
   Score a candidate even when it has never occurred with the query items.
3. **Test the value of combining both**
   Compare with each source collection alone and with published neural results.

Neural baselines

Architecture	Parameters	Architecture summary
CLHE	9.48M trained (POG)	Hierarchical attention with item- and bundle-level contrastive learning.
RaMen	16.06M trained (POG)	Characteristic and graph encoders combined with a learned hypergraph.
Bundle-MLLM	7B backbone	Multimodal item tokens projected into an LLM, adapted with LoRA.

BundleFuse

1. **Parameter-free scoring core**
   Zero gradient-trained scoring parameters.

2. **No neural-network training**
   Compute statistics and representations; select scalar settings on validation.

3. **Historical bundles as completion examples**
   Preserve joint evidence, transfer it through shared relations, and adjust reliance using candidate history.

General scoring equations

Final score: mix all selected scores with a specified subset.

$$\theta(c \mid S) = \frac{f_c}{f_c + \kappa} Y_c + \frac{\kappa}{f_c + \kappa} s_0 P_c$$

Y_c	Combination of all selected score sources
P_c	Category, descriptor or metadata scores, plus selected popularity; the chosen sources depend on the dataset
f_c	Candidate's number of appearances in training bundles
s_0, κ	Fallback scale and frequency threshold, respectively

At $f_c = 0$, only the scaled subset score $s_0 P_c$ remains.

General scoring equations

Combine standardized sources z_h with scalar weights.

$$Y_c = \sum_{h \in \mathcal{H}} w_h z_h(c \mid S), \qquad P_c = \sum_{h \in \mathcal{P}} \widetilde{w}_h z_h(c \mid S), \quad \mathcal{P} \subseteq \mathcal{H}.$$

$\mathcal{H}$: available sources. $\mathcal{P}$: chosen fallback sources.

$z_h(c \mid S)$: standardized score from source h.

$w_h, \widetilde{w}_h$: scalar weights selected for the two combinations.

The fallback combines selected sources already available to the full score.

General scoring equations

Put different score sources on a comparable scale.

$$z_h(c \mid S) = \frac{r_h(c \mid S) - \mu_h(S)}{\sigma_h(S) + \epsilon}$$

r_h	Raw score computed from history and available attributes
$\mu_h(S)$	Mean of this score over catalogue items, for query S
$\sigma_h(S)$	Standard deviation over the same items
$\epsilon > 0$	Small constant for numerical stability

No hidden completion labels enter these normalization statistics.

Ranking metrics at 20

Let G be the withheld items and L_{20} the top 20 recommendations.

$$\boxed{\text{Recall@20} = \frac{|L_{20} \cap G|}{|G|}}$$

Recall: fraction of withheld items recovered in the top 20.

Let $\text{rel}_r = 1$ if rank r is withheld, and 0 otherwise.

$$\text{DCG@20} = \sum_{r=1}^{20} \frac{\text{rel}_r}{\log_2(r+1)}, \qquad \boxed{\text{NDCG@20} = \frac{\text{DCG@20}}{\text{IDCG@20}}}$$

NDCG: rewards withheld items appearing earlier in the ranking.

IDCG@20 is the DCG of the ideal ordering. Both metrics are averaged over queries.

Results: POG

Full-catalogue ranking on the supplied split

Method	Recall@20	NDCG@20
CLHE (published)	0.02840	0.01930
RaMen (published)	0.03750	0.02260
BundleFuse	**0.05446**	**0.03609**

+45.2% relative Recall@20 versus published RaMen

Results: Spotify

Full-catalogue ranking on the supplied split

Method	Recall@20	NDCG@20
CLHE (published)	0.10810	0.18060
RaMen (published)	0.10910	0.18820
BundleFuse	**0.15565**	**0.25793**

+42.7% relative Recall@20 versus published RaMen

Results: why combine the evidence?

NDCG@20 on the same queries within each dataset

Evidence	POG	POG_DENSE	Spotify
Historical only	0.02094	0.25945	0.21170
Shared only	0.02929	0.07697	0.17191
Combined, no fallback	0.03394	0.27689	0.24320
Full BundleFuse	**0.03609**	**0.28788**	**0.25793**

Neither source family alone matches the combined estimator.

POG_DENSE full-model Recall@20: **0.39522**.